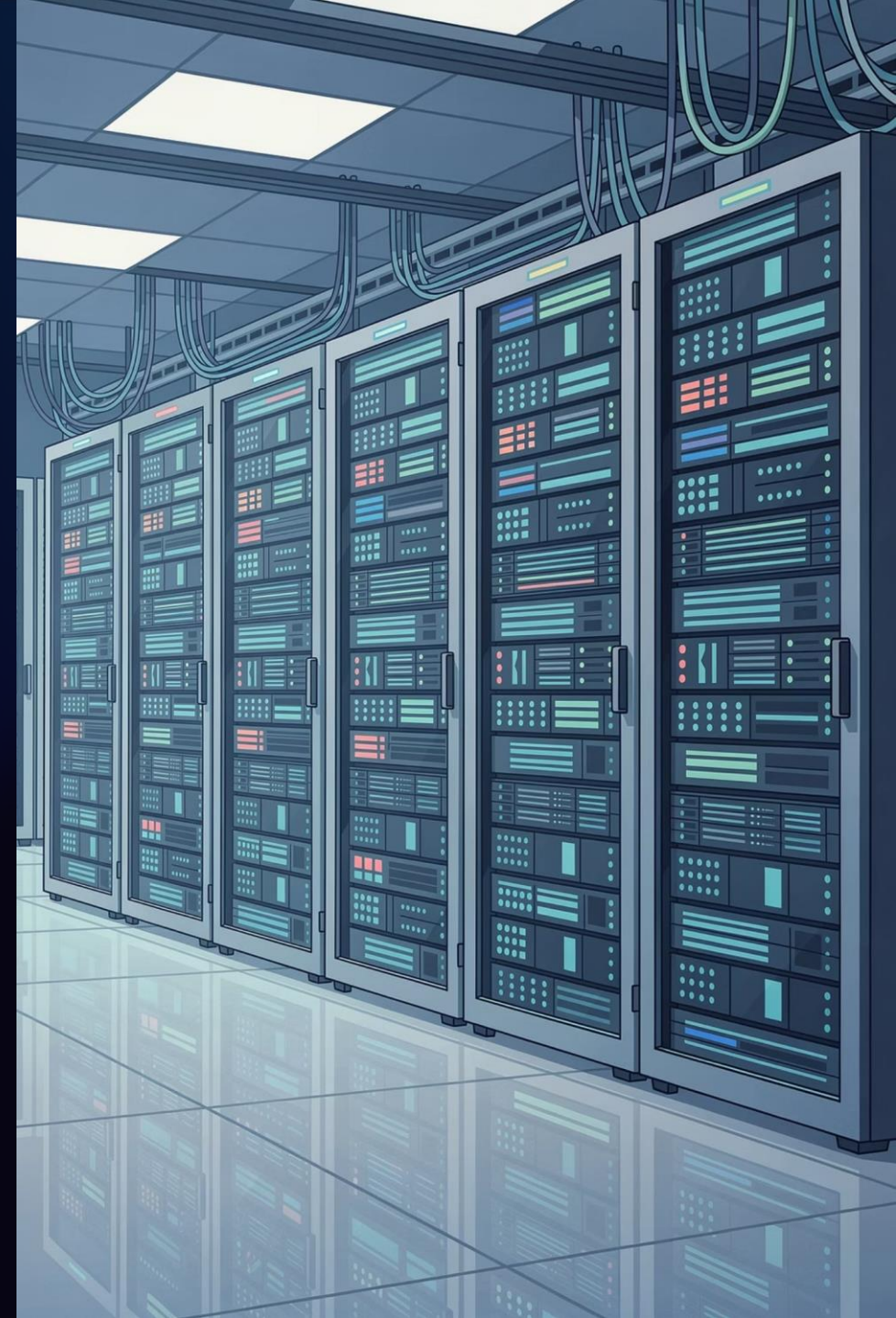


AWS Academy Cloud Foundations

Step-by-Step Hands-On Lab Guide

A practical guide for launching, scaling, and securing cloud infrastructure using the AWS Management Console. This guide covers Labs 3 and 6 — from spinning up your first EC2 instance to building a fully load-balanced, auto-scaling architecture.



Before You Begin

Critical Environment Rules

Do NOT Create Custom IAM Users or Groups

Attempting to create custom IAM Users, custom IAM Groups, or custom root billing structures will **instantly trigger an "Access Denied" error** due to AWS Academy sandbox restrictions.

Use Pre-Provisioned Resources

Always leverage the **pre-provisioned VPC environment** and any pre-existing security frameworks configured by the AWS Academy lab workspace. Do not attempt to recreate what already exists.

Verify Your Region

Always confirm the region dropdown in the top-right corner reads **N. Virginia (us-east-1)** before taking any action. Working in the wrong region is a common source of missing resources and errors.

LAB OVERVIEW

Two Labs, One Architecture Story

Lab 3 – Introduction to Amazon EC2

Launch, configure, secure, monitor, and resize a virtual Linux web server using the AWS Management Console. Learn the fundamentals of compute in the cloud.

Lab 6 – Scale and Load Balance Your Architecture

Build a horizontally scalable architecture using an AMI, an Application Load Balancer, and an Auto Scaling Group tied to dynamic CloudWatch alarms.



LAB 3

Lab 3: Introduction to Amazon EC2

Objective

Launch, configure, secure, monitor, and resize a virtual Linux web server using the AWS Management Console. You will experience the full EC2 instance lifecycle from launch to hardware modification.

What You Will Build

- A named EC2 instance running Amazon Linux 2023
- A custom security group with controlled HTTP access
- An Apache web server auto-configured via a boot script
- A resized instance with upgraded storage and type

Lab 3 – Getting Started

Prerequisites & Initial Setup

01

Log In to AWS Academy

Access your AWS Academy classroom account using your provided credentials. Navigate to the correct course and lab module.

03

Wait for Green Status

Wait until the status circle next to the AWS link turns **green**. This confirms your sandbox is fully ready. Then click the **AWS** link to open the Management Console in a new browser tab.

02

Click "Start Lab"

Click the **Start Lab** button at the top of the instruction panel. This provisions your personal sandbox environment with all required base resources.

04

Confirm Region: us-east-1

Check the region dropdown in the top-right corner. It must display **N. Virginia (us-east-1)** before proceeding. If it shows a different region, click it and select the correct one.

Launch an EC2 Instance

Navigate to the EC2 service using the top search bar, click **Launch instance**, and configure the following settings step by step.

1

Name & Tags

Under Name, type exactly: **Web Server** (case-sensitive)

2

AMI Selection

Keep default **Amazon Linux**. Confirm **Amazon Linux 2023 AMI** is selected from the dropdown.

3

Instance Type

Keep the default **t2.micro** (1 vCPU, 1 GiB memory). No change needed.

4

Key Pair

From the Key pair name dropdown, select **vockey**. This is the pre-generated key for Academy labs.

Lab 3 · Section 1 (Continued)

Configuring Network Settings

Click **Edit** next to Network settings and configure the VPC, subnet, and security group carefully. This step controls how your instance communicates with the outside world.

VPC

Select **Lab VPC** from the dropdown. Do not use the default VPC.

Subnet

Leave as the default **PublicSubnet1**. Auto-assign public IP must show **Enable**.

Security Group

Select **Create security group**. Name it exactly: **Web Server security group**. Description: *Security group for my web server*.

Remove Default Rule

Locate the pre-populated default SSH rule in the inbound rules section and click **Remove**. The group should start **completely locked down**.

Lab 3 · Section 1 (Continued)

Advanced Details & Boot Script

Storage Configuration

Keep the default **8 GiB gp3** root volume under Configure storage. No changes needed at this stage.

Termination Protection

Expand the **Advanced details** drawer. Find **Termination protection** and change it to **Enable**. This prevents accidental deletion of your instance.

User Data Script

Scroll to the bottom of Advanced details and paste this boot script into the **User data** field. It will automatically install and start Apache on first boot:

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo 'Hello From Your Web Server!' \
> /var/www/html/index.html
```

When finished, click **Launch instance** in the right Summary panel, then click **View all instances**.

Lab 3 · Section 2

Validate Server Health & Connectivity

01

Wait for Running State

In the Instances table, wait until Web Server transitions from *Pending* to **Running** and the **Status checks** column shows **2/2 checks passed**.

03

Test the Browser Connection

Open a new tab, paste the IP address, and press Enter. **Expected result:** the page will time out. This is intentional – the security group is blocking port 80.

02

Copy the Public IP

Click the checkbox next to Web Server. In the **Details** tab below, locate and copy the **Public IPv4 address**.

04

Open Port 80 via Inbound Rule

Go to **Security Groups** in the left nav, select Web Server security group, click **Edit inbound rules** → **Add rule**. Set Type: **HTTP**, Source: **Anywhere-IPv4 (0.0.0.0/0)**. Save rules.

- ✔ After saving the inbound HTTP rule, return to your browser tab with the Public IP and refresh. You should see: **"Hello From Your Web Server!"**

Lab 3 • Section 2 — Result

Your Web Server Is Live 🎉

What Just Happened?

The User Data script installed Apache and placed a static HTML page on your instance during boot. Opening port 80 allowed HTTP traffic to reach that page from any IP address on the internet.

Why the Page Timed Out Initially

Your custom security group had **no inbound rules** — this is a best-practice security posture called "deny by default." Traffic is only allowed once an explicit rule is added. This demonstrates the principle of least privilege.

Resizing & Modifying Hardware Configurations

EC2 instances can be resized — but only while stopped. Follow these steps to upgrade the instance type and expand the root volume.

1

Stop the Instance

Select Web Server → **Instance state** → **Stop instance**. Wait for state: **Stopped**.

2

Change Instance Type

Actions → **Instance settings** → **Change instance type**. Select **t2.small** and click Apply.

3

Enable Stop Protection

Actions → **Instance settings** → **Change stop protection**. Select Enable and Save.

4

Expand Storage to 10 GiB

Storage tab → click **Volume ID** → Actions → **Modify volume**. Change 8 → **10 GiB**, click Modify and confirm.

After modifications, navigate back to Instances, select Web Server, and click **Start instance**.

Lab 3 • Validation

Confirm Your Changes Took Effect



Instance Type

In the Instances table, confirm the **Instance type** column now shows **t2.small**. If it still shows t2.micro, the change was not applied — re-check that the instance was fully stopped first.



Root Volume Size

Click the **Storage** tab at the bottom of the instance details. Confirm the root volume now shows **10 GiB**. Modifications take effect within seconds of confirmation.



Termination Protection

Under the **Details** tab, verify that **Stop protection** is now enabled. This prevents the instance from being accidentally stopped via the console.

Lab 3 • Troubleshooting

Common Errors & How to Fix Them

● Browser Keeps Loading / Cannot Connect

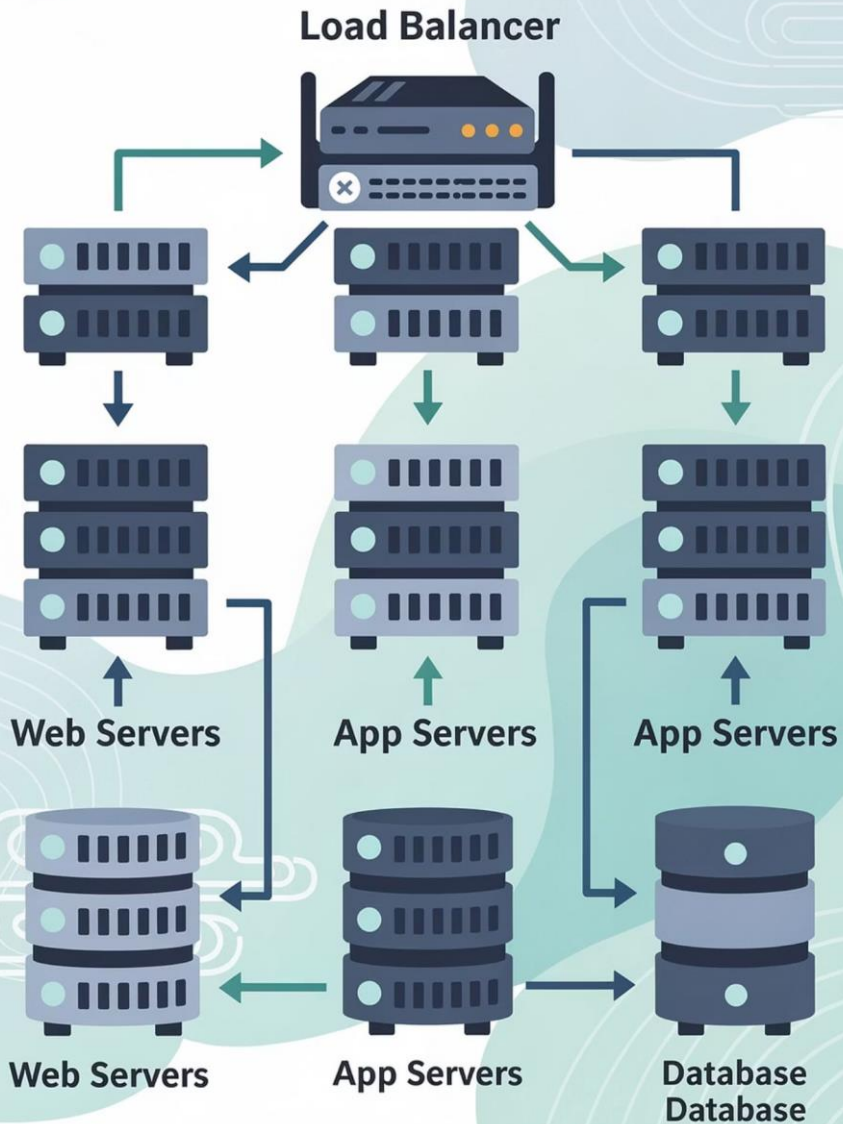
Cause: The security group inbound rule is missing or misconfigured.

Fix: Verify the rule type is explicitly **HTTP (Port 80)** and the source is **0.0.0.0/0**. HTTPS (Port 443) will not work for this lab — it must be HTTP.

● State Changes or Resizing Options Are Greyed Out

Cause: You are trying to modify the instance while it is still in a *Running* state.

Fix: You must first perform a full **Stop** of the instance and wait for the state to reach **Stopped** before changing the instance type or storage settings.

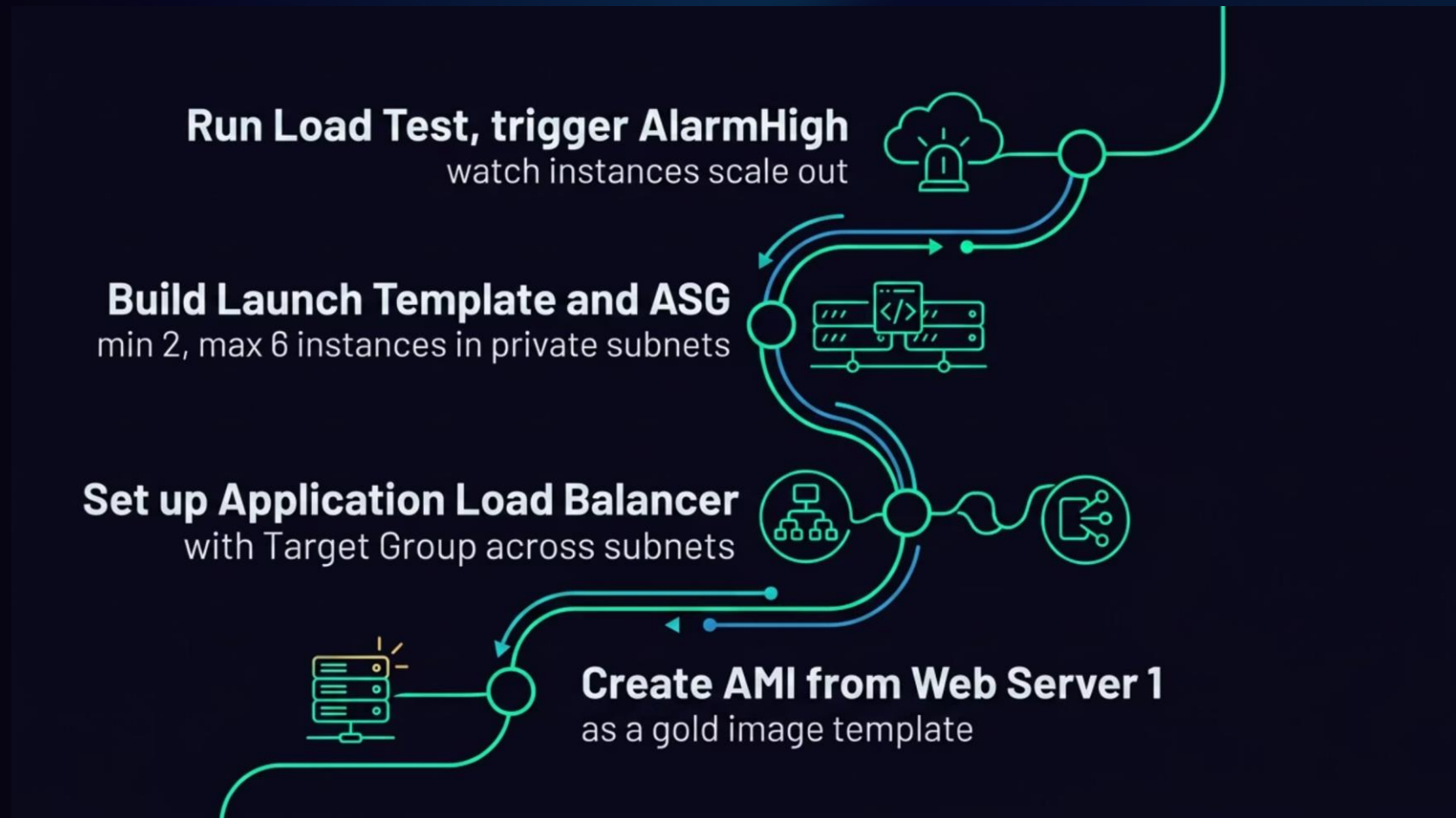


LAB 6

Lab 6: Scale and Load Balance Your Architecture

Move beyond a single server. In this lab, you will build a resilient, horizontally scalable cloud architecture using AMIs, an Application Load Balancer, and an Auto Scaling Group – all wired together with CloudWatch alarms.

What You Will Build



Each stage builds on the previous one, culminating in a fully automated, self-healing web architecture that responds to real CPU demand.

Lab 6 – Prerequisites

Getting Started with Lab 6

Log In and Start the Lab

Log in to your AWS Academy classroom and navigate to **Lab 6**. Click **Start Lab** and wait for the green status indicator before opening the AWS Console link.

Verify Region

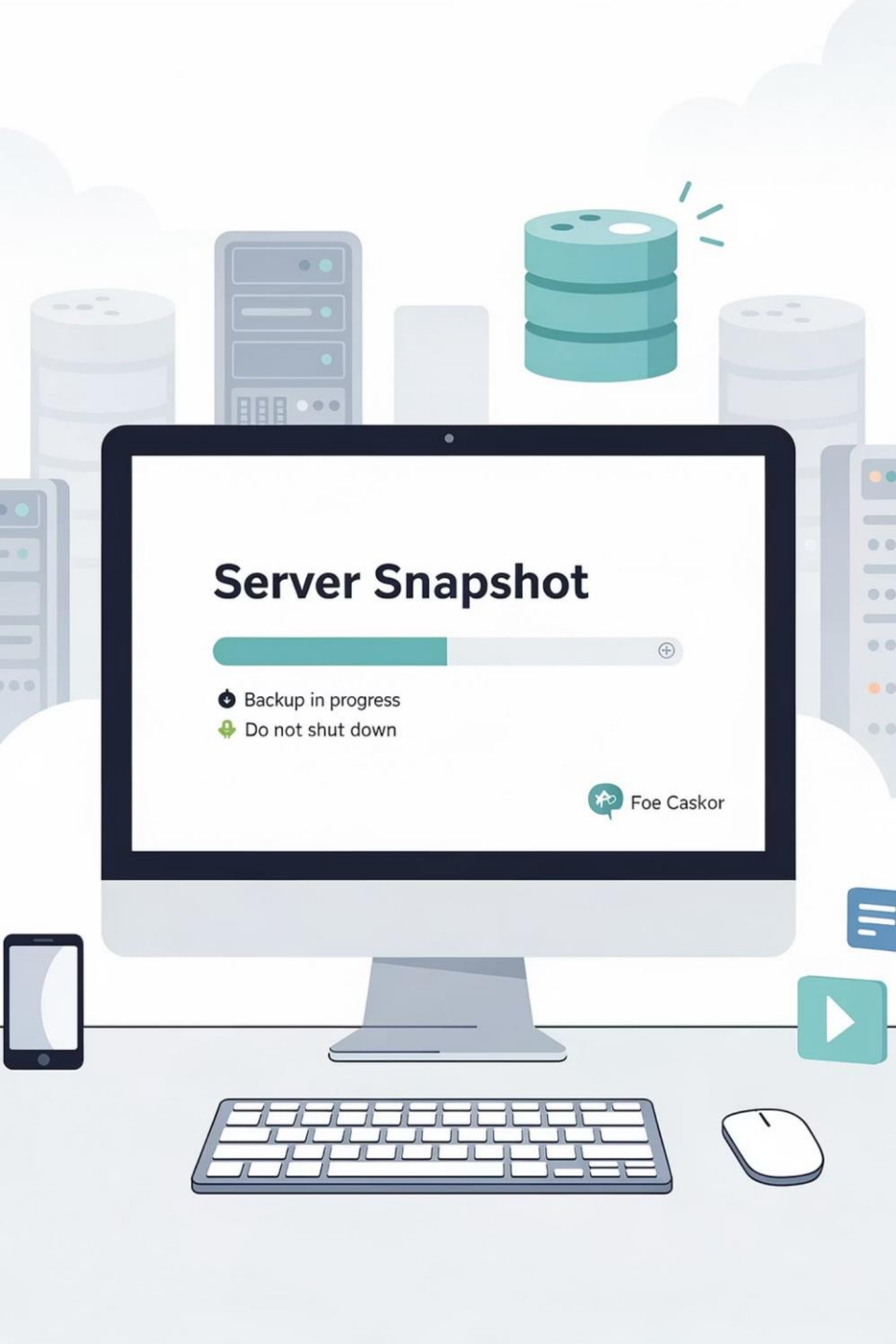
Confirm the region is set to **N. Virginia (us-east-1)** before proceeding.

Pre-Built Resource to Verify

Before building anything, confirm that an EC2 instance named **Web Server 1** is already running in your sandbox environment. This pre-built server will be used as the source for your AMI in the first section.



If Web Server 1 is not visible, wait a few more minutes for the lab environment to fully provision, then refresh the EC2 Instances page.



LAB 6 · SECTION 1

Create an AMI — Your Gold Image Template

An Amazon Machine Image (AMI) captures the full state of your server — OS, software, and configuration — as a reusable template for launching identical instances at scale.

01

Select Web Server 1

Go to EC2 Console → Instances. Click the checkbox next to the existing **Web Server 1** instance.

02

Initiate Image Creation

Click **Actions** → **Image and templates** → **Create image**.

03

Name the Image

Image name: WebServerAMI

Image description: Lab AMI for Web Server

04

Create the AMI

Click **Create image**. The AMI will appear under EC2 → AMIs with a status of *Pending*, then **Available**.

Lab 6 • Section 2

Create the Target Group

A Target Group tells the load balancer which EC2 instances should receive incoming traffic. You will register instances dynamically via Auto Scaling — so do not manually add any instances here.

1 Navigate to Target Groups

In the EC2 left nav, scroll to **Load Balancing** and select **Target Groups**. Click **Create target group**.

2 Configure the Group

Choose a target type: **Instances**. Set target group name to exactly **LabGroup**. For VPC, clear defaults and select **Lab VPC**. Click **Next**.

3 Skip Manual Registration

On the *Register targets* screen, **do not** manually check any active instances. Instances will be registered automatically by the Auto Scaling Group. Click **Create target group**.

Create the Application Load Balancer

1

Create ALB

Left nav → **Load Balancers** → **Create load balancer** → Under Application Load Balancer, click **Create**. Name it exactly: **LabELB**.

2

Network Mapping

VPC: **Lab VPC**. Mappings: Select **both Availability Zones**, mapping the first to **Public Subnet 1** and the second to **Public Subnet 2**.

3

Security Groups

Select **Web Security Group** from the dropdown. Click the **X** next to the default security group to remove it. Only Web Security Group should remain.

4

Listener Routing

For Listener **HTTP:80**, set the Default action to **Forward to LabGroup**. Click **Create load balancer**, then **View load balancer**.

Lab 6 · Section 3

Create the Launch Template

The Launch Template (LabConfig) defines the exact configuration Auto Scaling uses to spin up new instances — AMI, instance type, security group, and monitoring settings.

Core Settings

- **Name:** LabConfig
- **Auto Scaling guidance:** Check the box "Provide guidance..."
- **AMI:** My AMIs tab → select **WebServerAMI**
- **Instance type:** t2.micro
- **Key pair:** vockey
- **Security group:** Select existing → **Web Security Group**

Critical: Enable Detailed Monitoring

Expand **Advanced details**. Find **Detailed CloudWatch monitoring** and switch it to **Enable**.

This enables **1-minute metric intervals**. Without it, CloudWatch uses 5-minute basic intervals — your scaling alarms will be delayed and the load test may appear to not work.

Click **Create launch template** to finalize.

Lab 6 · Section 3 (Continued)

Create the Auto Scaling Group

From your newly created LabConfig template, open **Actions** → **Create Auto Scaling group**. Follow the wizard steps below.

01

Step 1: Name the Group

Name: **Lab Auto Scaling Group**. Click Next.

03

Step 3: Attach Load Balancer

Select **Attach to an existing load balancer** → choose **LabGroup**.
Check **Enable group metrics collection within CloudWatch**. Click Next.

02

Step 2: Network Configuration

VPC: **Lab VPC**. Subnets: Select both **Private Subnet 1** and **Private Subnet 2**. Click Next.

04

Step 4: Group Size & Scaling Policy

Desired: **2**, Min: **2**, Max: **6**. Scaling policy: **Target tracking**, policy name: **LabScalingPolicy**, metric: **Average CPU Utilization**, target: **60**. Click Next through remaining steps, then **Create Auto Scaling group**.

Auto Scaling Group Configuration Summary

2

Desired Capacity

Instances always running at baseline

2

Minimum

Floor: never drops below 2 instances

6

Maximum

Ceiling: scales out up to 6 instances

60%

CPU Target

Trigger threshold for LabScalingPolicy

Lab 6 • Architecture Design Note

Public vs. Private Subnets — Why It Matters

Application Load Balancer → Public Subnets

The ALB is internet-facing. It must live in **public subnets** so it can receive inbound traffic from the internet on Port 80. Public Subnet 1 and Public Subnet 2 are mapped in the ALB's network configuration.

Auto Scaling Group → Private Subnets

Your EC2 instances launched by the ASG live in **private subnets**. They are shielded from direct internet access. All traffic flows through the load balancer, which acts as the single entry point — improving both security and availability.

 Mixing up public and private subnets is the most common configuration error in Lab 6. The ALB goes in public subnets; the ASG instances go in private subnets.

Lab 6 • Section 4

Execute the Load Test

With your architecture fully deployed, it's time to stress-test it and watch Auto Scaling respond in real time.

01

Observe Initial Instances

Navigate to **EC2 Instances**. You will see 2 new servers named **Lab Instance** automatically initializing — these were launched by the Auto Scaling Group.

03

Open the Application

Paste the DNS string into a new browser tab. The web application should load successfully, confirming the ALB is routing traffic correctly.

02

Copy the ALB DNS Name

Go to **Load Balancers**, select **LabELB**, and copy the global **DNS name** from the details pane at the bottom.

04

Click "Load Test"

On the application page, click the **Load Test** button near the top header. Leave this tab **open** — it continuously generates artificial CPU load on your instances.

Lab 6 · Section 4 (Continued)

Watch CloudWatch Trigger the Scale-Out

Monitor the CloudWatch Alarm

Search for **CloudWatch** in the AWS console search bar. In the sidebar, click **All alarms**. Locate the alarm labeled **AlarmHigh**. Within a few minutes, the CPU utilization chart will breach the **60% threshold** and the alarm state will change to **In alarm** (shown in red).

Return to EC2 — New Instances Appear

Navigate back to **EC2 Instances**. You will observe **additional compute instances** automatically booting up. The Auto Scaling Group is distributing the heavy load across more servers — exactly as designed. The instance count will scale toward the maximum of **6 instances** as long as CPU remains above target.

Lab 6 • Validation

Confirm the Architecture Is Working



ALB DNS Loads the App

Paste the copied ALB DNS endpoint into your browser and confirm the **sample web application loads correctly**. This proves the load balancer is routing requests to healthy instances behind it.



Instance Count Scales Out

During the load test, validate that your **active instance count rises above the baseline of 2**, up toward the maximum of 6 – confirming that Auto Scaling is responding to the CloudWatch CPU alarm.



AlarmHigh State Triggers

In CloudWatch → All alarms, confirm **AlarmHigh** transitions to the **In alarm** state once CPU crosses 60%. This is the automated trigger that initiates the scale-out event.

Lab 6 • Troubleshooting

Common Errors & How to Fix Them

● Instances Show "Unhealthy" in Target Group

Cause: Public and private subnets have been mixed up in the configuration.

Fix: Verify that the **Application Load Balancer** uses **Public Subnets** and the **Auto Scaling Group** uses **Private Subnets**. Mixing these layers blocks internal health check probes from the ALB reaching the instances.

● Auto Scaling Group Won't Scale Out During the Test

Cause: **Detailed CloudWatch monitoring** was not enabled in the Launch Template.

Fix: Without it, CloudWatch uses **5-minute basic intervals** — alarms may take much longer to trigger. Re-open the Launch Template, edit it, expand Advanced details, and enable **Detailed CloudWatch monitoring**.

Concept Review

Key AWS Services Used in These Labs



Amazon EC2

Elastic Compute Cloud provides resizable virtual servers. You control OS, instance type, storage, and networking — the foundational compute building block.



Security Groups

Virtual firewalls that control inbound and outbound traffic at the instance level. Rules are stateful — allowing a request also allows the response.



Application Load Balancer

Routes HTTP/HTTPS traffic to healthy instances across multiple Availability Zones. Provides fault tolerance and even traffic distribution.



Auto Scaling Group

Automatically adjusts the number of EC2 instances based on demand. Uses Launch Templates to define what each new instance looks like when it boots.

Concept Review (Continued)

Supporting Services: AMI & CloudWatch

Amazon Machine Image (AMI)

An AMI is a **snapshot of a configured server** — including the operating system, application software, and settings. It acts as a "gold image" template that Auto Scaling uses to launch new instances that are pre-configured and deployment-ready. In Lab 6, **WebServerAMI** was your gold image.

Amazon CloudWatch

CloudWatch is AWS's monitoring and observability service. It collects metrics like **CPU utilization**, sets alarms when thresholds are breached, and can trigger automated actions — like scaling out an Auto Scaling Group. **Detailed monitoring** (1-minute intervals) is critical for responsive scaling behavior in lab environments.

Lab Completion

You've Completed Both Labs

Lab 3 Achievements

- Launched an EC2 instance with a boot script
- Configured a security group to allow HTTP traffic
- Resized instance type from t2.micro → t2.small
- Expanded root volume from 8 GiB → 10 GiB

Lab 6 Achievements

- Created a reusable AMI from a running server
- Built an Application Load Balancer with Target Group
- Configured a Launch Template with detailed monitoring
- Deployed an Auto Scaling Group across private subnets
- Triggered and observed a real-time scale-out event

Remember: Always click **End Lab** in your AWS Academy panel when finished to release sandbox resources and avoid timeouts.

